

SceneLLM: Dynamic Scene Graph Generation via Video-to-Language Mapping and Large Language Models

SceneLLM (2024) presents a framework for generating **dynamic scene graphs** from video by using Large Language Models (LLMs) to perform **temporal and relational reasoning** across visual frames. The core idea is to convert videos into language-like sequences and use LLMs to infer evolving object relationships over time — making SceneLLM a dynamic graph reasoning system applied to visual data.

What is a Dynamic Scene Graph?

A **scene graph** represents a single video frame as a graph:

- **Nodes** = objects (e.g., person, cup, chair)
- **Edges** = relationships between objects (e.g., holding, sitting on)

In videos, objects move and interact differently over time. Thus, scene graphs **change frame by frame**, forming a **dynamic graph**:

- Frame 1: <person, holding, cup>
- Frame 2: <person, drinking from, cup>
- Frame 3: <person, dropped, cup>

These evolving triplets represent dynamic edges between object nodes.

How SceneLLM Works

SceneLLM operates in three main stages:

1. Video-to-Language (V2L) Mapping

- Extracts object features and motion info using a video encoder (e.g., EVA-CLIP)
- Converts this into structured language prompts describing the scene (e.g., “Person A is near a cup on a table.”)

2. LLM-based Temporal Reasoning

- Feeds these prompts into a frozen LLM (e.g., LLaMA-2) to model spatio-temporal context
- Outputs embeddings that encode how object interactions evolve across frames

3. Scene Triplet Prediction

- A decoder predicts semantic triplets like `<dog, behind, sofa>` or `<person, sitting on, chair>`
- These triplets form the edges in the output scene graph

The process is repeated across frames to generate a **sequence of scene graphs**, capturing temporal object dynamics.

Results and Evaluation

SceneLLM is evaluated on two major video SGG benchmarks:

- **VidOR** (Video Object Relationship)
- **Action Genome**

Key findings:

- SceneLLM outperforms previous models on **triplet prediction accuracy**
- Especially effective in **long video sequences** and **low-data regimes**
- Demonstrates strong generalization due to the LLM's language understanding capabilities

Why It Matters for the Survey

SceneLLM contributes a unique multimodal example to the field of **LLM-based dynamic graph reasoning**:

- Applies LLMs not just to text or tabular data, but to **visual-temporal input**
- Models **dynamic graphs where both structure and semantics evolve over time**

- Integrates **vision-language fusion** and **symbolic graph prediction** through language reasoning